AI (behaviourism) and as a proposal for finding higher order function in physical structures (materialism). We argue then for the fact that none of them can provide a full blown account of agency and intelligence.

Our next step is to analyse the ps model, a form of utility-based agent recently developed in the field of embodied cognitive science. Its internal working structure as a stochastic network of so-called "clips" step-wise updated, allows us to individuate its main features: composition and the possibility to take future possible experiences as its main argument. We spot in its capability to perform projections its main characteristic.

We show that none of the two theory-driven approaches to AI is able to account for this feature, and we suggest that projection is a functional link, namely a distinct feature which we cannot account for according to the separate views, but which is, at the same time, necessary in order to bridge together behaviourism and materialism. We show in fact that for a full blown characterisation of agency we must refer both to behaviouristic elements (the agent-environment interactions) and materialistic ones (internal processes which may not be manifest as a change in the agent's behavior). This idea provides a broader view of the concept of agency, since it allow us to keep projection, as a functional-link, at the centre of the scene.

This analysis allows us to present a feature-driven (or reversed) taxonomy of the concept of agency. Instead of suggesting a top-down approach, we individuate a central feature which guarantees a richer view of the problem. We sketch the main characteristics of our taxonomy and we show that it allows a comparison of different agents, based on the individuated functional-link of projection, which is richer than the solely behaviouristic and materialistic approaches. The reason for that lies precisely in the fact that we have reversed the approach on agency from a theory-driven stance to a process-driven one.

105 Conscienceness – the correlation between Machine Ethics and Machine Consciousness

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This research's main focus is to highlight the correlation between Machine Ethics (ME) and Machine Consciousness (MC). Since the very beginning of Machine Ethics as a field of science authors like Michael and Susan Leigh Anderson argued that "from family cars that drive themselves (...) to fully autonomous robotic entities (...) it is clear that machines such as these will be capable of causing harm to human beings unless this is prevented by adding an ethical component to them". This statement supports "Frankenstein complex", the human fear of creation bringing destruction down upon its creator. One hundred years ago in Karel Capek's renewal of this idea in his "R.U.R." we found the word "robot" which got immediately linked to a hostile robot rebellion leading to the extinction of the human race. Picked up by Isaac Asimov, the idea developed into creation of The Three Laws of Robotics. But was this invention really needed? Would not a "do not crash" rule be enough for an AI-driven car instead of a "do not crash because it is wrong" one? In my approach I call it an "AI Ockham razor" approach which I define as "omitting AI-based solution when- and wherever it is possible". What does not imply that we should withdraw developing AI research. Humans morality operates on two basic level: moral intelligence allowing us to differ good from wrong and moral competence making us do the right thing despite the circumstances. For an Artificial Moral Agent it will always be a socratic approach: to know good is to do good. That is why Machine Ethics research focuses on the former. On the other hand, there actually may never be Machine "Ethics" as long as Machine Consciousness research gets left behind and at least a poor substitute for a free will is not provided. The question is, should we really pursue creating a potentially unethical Agent? The answer – not so simple. This research advocates MC concept based on a triple awareness model consisting of: environment-awareness in both passive (being an object to interact with) and active (having an impact on other objects, e.g. pushing a mug of a shelf) way, a finitude awareness including categories of damaging (a broken mug handle), destroying (a mug shattered into pieces) and killing a thing or a person, and an objects' hierarchy scale awareness (putting agent's existence above mug's but never – human's). All of these factors taken by the Agent into consideration should make its environment functioning and HCI experience more natural and substantive, and that is this approach's main goal.